CardioVision AI

TRACK A • READY TO SUBMIT

Multimodal AI Hackathon 2026 — Devpost Submission Master Kit

1. WHAT'S YOUR PROJECT CALLED?

Exact match for Project Title field

CardioVision AI

2. ELEVATOR PITCH

167 / 200 characters

An interactive 3D cardiovascular risk platform predicting disease risk from clinical data and mapping calibrated AI insights onto a rotatable 3D anatomical human heart.

3. BUILT WITH (TAGS)

Add as comma-separated tags in Devpost

`python` `react` `three.js` `webgl` `lightgbm` `scikit-learn` `fastapi` `flask` `vite` `tailwindcss` `pandas` `numpy` `shap` `pythonanywhere`

4. "TRY IT OUT" LINKS

Click "Add another link" to include both

🌐 Live Web Application:	https://keerthana08.pythonanywhere.com
💻 GitHub Repository:	https://github.com/KEERTHANAHL08/cardiovision-ai

5. MEDIA, THUMBNAIL & VIDEO DEMO

Visual Assets

- **Thumbnail Image (3:2 ratio):** Upload `frontend/src/assets/hero.png` from your project.
- **Gallery Images:** Upload screenshots of (1) 3D Heart with margin callouts, (2) Assessment Form with Demo Cards, (3) AI Insights Feature Contribution charts.
- **Video Demo Link:** 1–2 minute screen walkthrough of the live web app uploaded to YouTube or Vimeo.

6. ABOUT THE PROJECT (STORY MARKDOWN)

Copy and paste directly into the Devpost "About the project" Markdown editor

💡 Inspiration

Cardiovascular disease (CVD) remains the leading cause of mortality worldwide, yet clinical risk scores are often presented as abstract statistical percentages that patients struggle to comprehend. We wanted to bridge the gap between complex machine learning and intuitive patient communication by creating an interactive, multimodal bridge: translating 12 physiological markers directly into an interactive, rotatable 3D anatomical heart visualization.

🫀 What It Does

CardioVision AI is a clinical-grade cardiovascular risk prediction and 3D visualization platform:

- **Calibrated Risk Prediction:** Evaluates patient vitals (systolic/diastolic blood pressure, cholesterol, glucose, BMI, pulse pressure, smoking, and activity levels) to output an empirical CVD risk probability (0% – 100%).
- **Encyclopaedia Britannica 3D Heart Engine:** Renders a high-fidelity 3D human heart (`heart.glb`) in WebGL with continuous 360° rotation and interactive OrbitControls (pan, zoom, tilt).
- **Dynamic Margin Callouts & SVG Leader Lines:** Pinned callouts on the screen margins are anchored directly to real 3D surface coordinates of 12 distinct anatomical structures via dynamic cubic Bezier leader lines.
- **Patient Vulnerability Mapping:** Connects specific physiological abnormalities (systolic hypertension, dyslipidemia, pulse pressure overload) to corresponding anatomical regions (Left Ventricle, Coronary Arteries, Aortic Arch).
- **Explainable AI (XAI):** Provides local feature attribution breakdowns showing exactly which clinical markers increase or decrease risk, complete with interactive counterfactual what-if simulation.

⚙️ How We Built It

- **Machine Learning Pipeline:** Trained and bench-marked 6 classifiers over 5-fold Stratified Cross-Validation on 68,637 clinical records (`cardio_train.csv`). The champion **LightGBM Classifier** achieved **0.8034 ROC-AUC**, **0.7289 F1-score**, and **73.82% Accuracy**, calibrated with Isotonic Calibration.
- **Feature Engineering:** Derived hemodynamic indicators including Body Mass Index (BMI), Pulse Pressure (arterial stiffness marker), and Mean Arterial Pressure (MAP).
- **Frontend Architecture:** Built with React 18, Vite, and Tailwind CSS.
- **3D Graphics Engine:** Developed with Three.js (WebGL), GLTFLoader, and screen-space unprojection algorithms (`Vector3.project(camera)`).
- **Full-Stack WSGI Deployment:** Unified into a high-performance Flask WSGI backend (`wsgi.py`) with vectorized batch inference and LRU caching, deployed live on PythonAnywhere.

🧗 Challenges We Ran Into

- **Overlapping 3D Billboards:** Traditional floating 3D labels cluttered the heart model when rotated. We solved this by engineering an *Encyclopaedia Britannica* plate layout with fixed margin columns and dynamic 2D-to-3D SVG Bezier leader lines that recalculate smoothly on every frame.

- **Cloud Latency Optimization on Shared CPU:** Initial unvectorized feature perturbation loops caused high latency on PythonAnywhere's free CPU tier. We restructured the computation into a single vectorized batch evaluation matrix and added in-memory LRU caching, slashing response latency from 12+ seconds to under **50 milliseconds**.
- **Clinical Truth & Calibration:** Standard tree models often produce uncalibrated probabilities. We implemented probability calibration (`CalibratedClassifierCV`) to ensure predicted risk deciles correspond to empirical population outcomes.

🏆 Accomplishments That We're Proud Of

- Designing a publication-grade 3D heart interface that feels like an interactive medical textbook.
- Achieving sub-50ms cloud inference with zero lag on a live public deployment.
- Creating an honest, responsible AI system that provides actionable counterfactual insights without making unsubstantiated clinical diagnoses.

📚 What We Learned

- Advanced 3D coordinate transformation and matrix projection in Three.js.
- Techniques for optimizing machine learning inference loops in constrained cloud environments.
- How to communicate complex ML probabilities effectively through visual medical design.

🔮 What's Next For CardioVision AI

- Integration with FHIR/HL7 electronic health record (EHR) standards.
- Longitudinal tracking to visualize how patient risk trajectories change over multi-year clinical checkups.
- Mobile WebXR/AR support to project the 3D heart onto a clinic desk using smartphones or AR headsets.